

Let’s Face It: How Markets React to the Fed Chair’s Facial Expressions

Based on Curti & Kazinnik (2023), Journal of Monetary Economics

ANDREW SANGWOO YE, HYEMIN YOON

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Do Markets Read the Federal Reserve Chair's Face?

- Central bank communication as a key monetary policy tool
- FOMC press conferences transmit words + visual cues
- Research Question: **Do the Fed Chair's facial expressions predict market reactions beyond the information in their words?**

Why Should a Facial Expression Matter?

1. Information asymmetry	Investors search for additional signals
2. Negativity bias	Negative signals receive disproportionate attention
3. Soft information	Facial expressions may reveal information about the economic outlook that words do not fully capture

From Video Frames to Market Data

Sample

- 46 press conferences
- 2011–2020
- 3 Fed Chairs
- 2,518 minute-level observations

Facial expressions

- Frames every 2 seconds
- Microsoft Azure, CNN-based emotion recognition
- Anger + Disgust + Fear, averaged over the prior 3 minutes and normalized by Chair

Market analysis

- Subsequent 3-minute market changes
- SPY, VIX, EUR/USD, SPY volume
- OLS regressions with verbal controls and fixed effects

Class connection: Computer vision turns visual data into quantitative research variables.

How a Facial Expression Becomes a Score



Panel C: Jerome Powell, January 30th 2019

Emotion	Intensity Score
Anger	0.00
Contempt	0.05
Disgust	0.00
Fear	0.00
Happiness	0.00
Neutral	0.04
Sadness	0.91
Surprise	0.00

Each frame receives eight emotion scores that sum to one

Combining Visual and Textual Measures

FinBERT

Measures textual sentiment:
Positive, negative, neutral

GPT-3 embeddings + classifier

Measures policy stance:
hawkish vs. dovish

Do facial expressions still matter, after controlling for what the Chair actually says?

Class connection:
Visual and textual channels are measured separately and combined in the analysis.

Negative Facial Expressions and Market Reactions

Estimated 3-minute changes for a 1 SD increase in negative expressions

S&P 500 (SPY)

−0.53 bp

Significant at 5%

Lower equity prices

Volatility (VIX)

+3.76 bp

Not statistically significant

Higher uncertainty

EUR/USD

−0.18 bp

Significant at 10%

Stronger USD

SPY volume declines: −12,702 shares per minute (−2.85%).

Market Reactions Are Stronger During Forward Guidance

- **Negative expressions have stronger associations** with market movements when the Chair discusses future policy.

- This pattern appears across SPY, VIX, and EUR/USD.

Are Facial Expressions Informative?

Maybe the Chair is simply tired, or stressed?

- **Evidence of information content:** Negative facial expressions correlate with the negative tone of FOMC minutes,
- **Limited evidence of learning:** The SPY response becomes stronger with Chair tenure, but the evidence is weak.

These patterns support, but do not establish, an informative channel.

How convincing is the Evidence?

Why we find it convincing

- Short windows reduce exposure to unrelated news
- Text controls help separate verbal and visual information
- SPY results persist across fixed-effect specifications

What remains uncertain

- Voice and other simultaneous cues may remain
- Only 3 chairs
- Scores depend on Azure

Why Does This Matter Beyond the Study?

For Research

- Combine facial and textual measures in other public communication settings.

For Central Banks

- Evaluate how communication formats affect interpretation and public transparency.

Ethical concern:
Emotion scores may be misread or perform unevenly across people.

Our Extension: Taking It to Korea

We applied the paper’s method to Bank of Korea press conferences. Why Korea?

- We are both Korean
- We both trade in the Korean market
- Every Bank of Korea press conference moves our portfolio
- Can we use this study?

Andrew

−14.8%

12-month portfolio return

Win rate: 31%

Hyemin

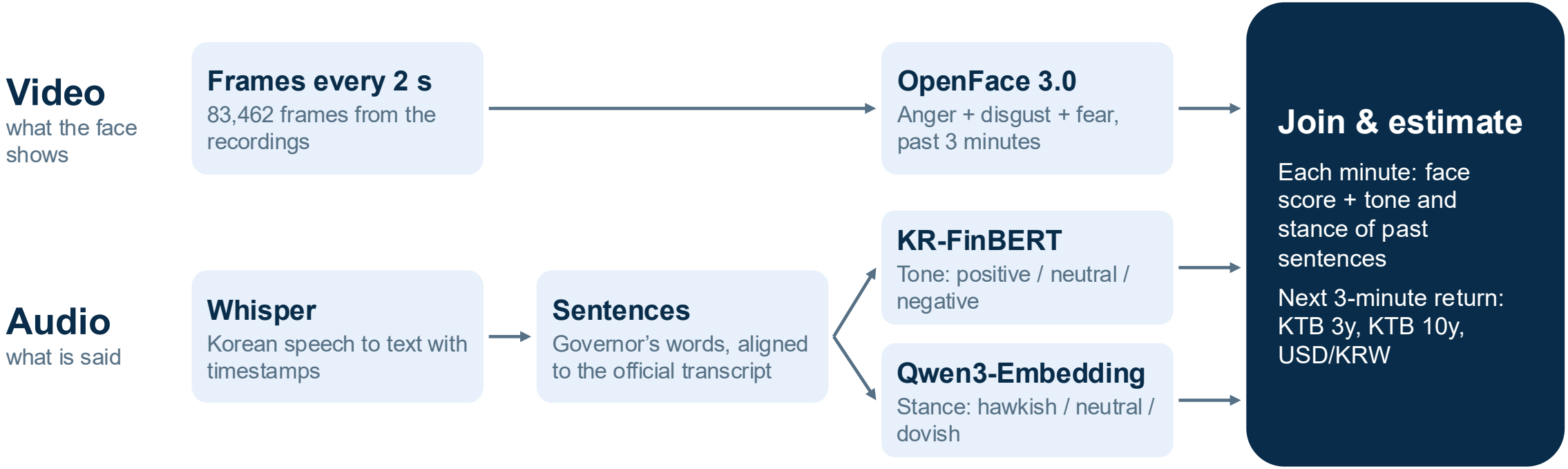
+41.3%

12-month portfolio return

Win rate: 62%

Our Extension: The Pipeline

33 Bank of Korea conferences, 2020–2026 · ≈46 hours of video · open-source models on ETH's Euler GPU cluster



What the Model Sees: Real Frames

Frames from our run with their OpenFace 3.0 scores · green box = detected face



KEPT · 2023-08-24, 300 s into video

Surprise	0.91
Contempt	0.04
Disgust	0.03
Fear	0.02

Negativity: 0.05



KEPT · 2020-10-14, 2400 s into video

Disgust	0.90
Sad	0.03
Contempt	0.02
Surprise	0.01

Negativity: 0.92



EXCLUDED · 2022-08-25, 300 s into video

Surgical mask covers the mouth

Mask periods are dropped, as are wide shots, reporters and other speakers.

Results: No Clear Association in Korea

Estimated 3-minute changes for a 1 SD increase in negative expressions

3-year bond futures

−0.24 bp

Not statistically significant

95% CI [−0.83, +0.34]
18 conferences

10-year bond futures

+0.70 bp

Not statistically significant

95% CI [−1.35, +2.76]
23 conferences

USD/KRW

+0.05 bp

Not statistically significant

95% CI [−0.63, +0.74]
33 conferences

Fed study: SPY −0.53 bp, significant at 5%. Adding our hawkish/dovish control barely moves the estimates (without it: −0.24, +0.74, +0.01 bp).

How Should We Read a Null Result?

Korea may genuinely differ

- Bond futures and FX, not equities
- Different market and audience
- Display norms may vary across cultures

Our test may miss an effect

- Fewer conferences: 33 vs. 46
- Models not validated on Korean faces

Absence of evidence is not evidence of absence: our estimates are inconclusive, not zero.

References

[1] Curti, F., & Kazinnik, S. (2023). *Let's face it: Quantifying the impact of nonverbal communication in FOMC press conferences.* Journal of Monetary Economics, 139, 110–126.
DOI: 10.1016/j.jmoneco.2023.06.007

[2] *Computational Social Science with Images and Audio* (2026). Lecture slides: Sessions 1a, 1b, 2a, and 2b. ETH Zurich. https://github.com/philinew/css_images_audio/tree/main/2026/slides

If AI can trade on facial expressions in real time,

Should Central banks keep **live press conferences**,

Or move toward **more controlled form of communication**?